D.O.P.E.: Downstream Objective-Preserving Encoding

Generative Compression of Predictive Structure

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Abstract

How much of a table's predictive structure can a small generative program carry? Downstream Objective-Preserving Encoding (D.O.P.E.) compiles a numeric regression table into a sampling program (quantized marginals, dependence model, target residual, noise model) under a 10,240-byte cap (median 1,895 bytes; $490/500$ fits within it); sampling reads only the artifact. We score retention, a null-normalized train-on-synthetic/train-on-real ratio, across diverse model families on 100 PMLB regression tables with five DOPE fit seeds and paired, family-clustered intervals. At size $4n$, DOPE's median retention is 1.133 [1.029, 1.246] under the neural MLP, 0.991 [0.972, 0.997] under linear regression, and 0.898 [0.804, 0.972] under CatBoost, alongside pretrained foundation models (Kumo, TabICL, Mitra) on the density block. DOPE cannot be separated from TabSyn (scaled schedule), the strongest full-panel generator (Hodges–Lehmann difference -0.003 [-0.053, 0.053]), whose 42.3 MB median artifact is 2.2×10^4 times DOPE's, and cannot be separated from author-default ARF (0.048 [-0.034, 0.149]). On a separate 6-lineage block, DOPE is below Forest-Flow (-0.231 [-0.382, -0.005]). It is above a predictor-only sampler (0.197 [0.112, 0.267]) and is below four-fold resampled real rows (-0.179 [-0.276, -0.074]). All estimates use validation rows that also compared DOPE's profile with three alternatives, so every lineage and auditor is selection-exposed; the official test split is reserved for one pre-registered run.

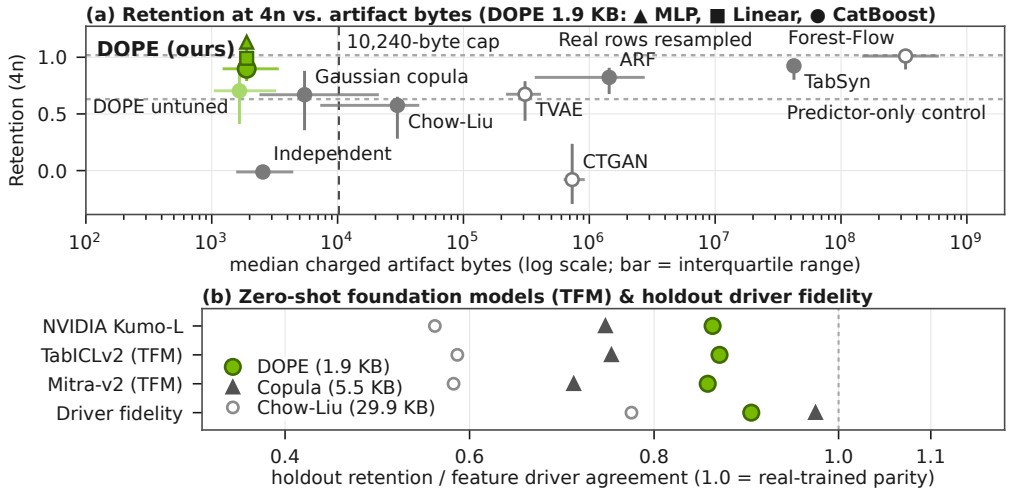


Figure 1: **What DOPE keeps, and at what size. Top (a):** Retention at $4n$ (median and 95% nested bootstrap interval) against median charged artifact bytes (log scale; horizontal bar: interquartile range), showing DOPE (1,895 bytes) across neural MLP, linear, and CatBoost inside the 10,240-byte cap (dashed line); dotted lines mark the predictor-only and four-fold real-row controls (a row-count diagnostic). **Bottom (b):** Zero-shot foundation models (NVIDIA Kumo large, TabICL, Mitra) and holdout driver fidelity (Tables 1 and 3; selection-exposed validation).

1 INTRODUCTION

An organization's tabular data encodes its operational experience and competitive moat, yet security, privacy, and legal constraints routinely prevent teams from exporting raw tables to test external machine learning architectures or pretrained tabular foundation models. A synthetic release solves that bottleneck only when a model trained or conditioned on synthetic rows transfers to unseen real rows with the same predictive lift and feature drivers as a model trained on the real table. We call what must survive *predictive structure*: the relationships between the features and the target that a declared learner uses, measured by the share of a real-trained learner's gain over a null that a learner trained only on synthetic rows recovers on real validation rows. The term is operational and relative to a workload, meaning a target, a learner, a synthetic size, and a loss; it names no general store of knowledge.

Compression is a natural way to pose the question. Copying or memorizing a dataset into a megabyte-scale checkpoint is trivial; packing the transferable predictive regularities into a tiny generative program requires separating signal from noise (Rissanen, 1978; Blumer et al., 1987), and the information bottleneck frames compression relative to a target variable (Tishby et al., 2000). We use that link as motivation only. An empty generator is tiny, a constant table compresses well, and an external checkpoint can hide most of a representation, so size alone shows nothing about learning. Downstream Objective-Preserving Encoding (D.O.P.E., or DOPE) therefore charges every byte that sampling reads and judges an artifact only by what learners trained on its samples recover; retention estimates neither mutual information nor description length.

Two practices make that property hard to compare across generators. First, released generators are large: CTGAN and TVAE (Xu et al., 2019), TabDDPM (Kotelnikov et al., 2023), TabSyn (Zhang et al., 2024), and forest flows (Jolicoeur-Martineau et al., 2024) ship hundreds of kilobytes to gigabytes of weights, which are expensive to audit, to store next to a dataset, and to reason about as a release. Second, a generator paper often reports a single downstream learner or the metric that also chose its configuration, so model generality and selection exposure stay entangled.

DOPE starts from the downstream task. When the intended use is regression on a target y, a generator only has to reproduce the feature distribution well enough and the conditional mean $\mathbb{E}[y \mid x]$ well enough for diverse learners to recover it. DOPE therefore stores the conditional mean explicitly, as a tiny residual network over a screened set of inputs, and stores the features as quantized marginals tied together by a compact dependence model. The whole artifact must fit a byte budget, and sampling reads nothing but the artifact.

We evaluate with one measurement across six model architectures. Retention (3) asks how much of a real-trained auditor's improvement over a null an auditor recovers when trained or conditioned on synthetic rows: three task-trained learners (neural MLP, linear regression, and CatBoost) on the full panel and three zero-shot pretrained tabular foundation models (NVIDIA Kumo large, TabICL, and AutoGluon Mitra) alongside holdout feature driver fidelity on the density block. Two controls separate the ingredients: resampling real rows four-fold isolates the effect of row count, and a predictor-only sampler, which keeps a fitted regressor but drops the joint feature model, isolates the dependence model. Every comparison is paired by lineage, uses five DOPE fit seeds on the main panel, and carries a nested bootstrap interval with simulated source families as clusters. Each method's selection data are stated, including DOPE's own profile comparison, which read validation rows of every lineage in this panel (Section 3.3).

Contributions.

1. **A generative program for predictive structure.** Compiled from the table, DOPE stores per-feature marginals and transforms, a dependence model chosen among seven families on the training rows, a twelve-input residual for the target, and a heteroscedastic noise model, all quantized into an artifact whose median charge is $1{,}895$ bytes (Section 3, Algorithm 1).
2. **Multi-model and foundation-model transfer.** Retention is credited as a TSTR/TRTR ratio across task-trained auditors (neural MLP, linear, CatBoost) at sizes n and $4n$ and zero-shot foundation models (Kumo large, TabICL, Mitra) on the density block, beside row-count and predictor-only controls and holdout driver fidelity (Section 4).
3. **Where DOPE stands across model families, and why.** At $4n$, DOPE retains 1.133 under the neural MLP, 0.991 under linear regression, and 0.898 under CatBoost; it cannot be separated

from TabSyn (scaled schedule) while storing 2.2×10^4 times fewer bytes, cannot be separated from author-default ARF, is above the predictor-only control, and is below resampled real rows, which diagnose row count rather than bound retention (Section 5).

4. **A release gate that fails closed.** MFS-v3 is a gated geometric mean with an executable conformance test. It is null for every method here, and the paper names the gates that keep it null (Section 3.4).

2 Related work

Generators. Copula and Bayesian-network models (Patki et al., 2016; Chow & Liu, 1968; Aas et al., 2009; Zhang et al., 2014) fit explicit, often small, joint models; marginal-based private synthesizers (McKenna et al., 2021; 2022; Cai et al., 2021) add formal privacy. GANs and VAEs (Xu et al., 2019; Park et al., 2018; Zhao et al., 2021), diffusion and score models (Kotelnikov et al., 2023; Zhang et al., 2024; Kim et al., 2023; Lee et al., 2023; Shi et al., 2025), forest flows (Watson et al., 2023; Jolicoeur-Martineau et al., 2024), language-model generators (Borisov et al., 2023; Solatorio & Dupriez, 2023), sequential CART (Nowok et al., 2016), and prior-fitted networks (Ma et al., 2023; Hollmann et al., 2025) optimize a likelihood, a denoising objective, or adversarial agreement on the joint table. DOPE optimizes a different thing: the target's conditional mean, explicitly, under a byte budget, with the joint feature model kept compact; supplement K tabulates each family against DOPE.

Compact and program models of tables. Bayesian synthesis of probabilistic programs (Saad et al., 2019) and CrossCat (Mansinghka et al., 2016) also produce small, inspectable generative models of tables. They target the joint distribution and posterior inference; DOPE targets a downstream regression objective and charges every artifact in bytes against a release cap.

Distillation, compression, and naming. Dataset distillation (Wang et al., 2018) learns a few synthetic training examples that preserve a model's accuracy, and knowledge distillation (Hinton et al., 2015) moves a trained model's behavior into a smaller model. Both keep learned behavior rather than records, as DOPE aims to, but they ship examples or weights; DOPE ships a program that draws fresh rows, and its predictor-only control tests what that program adds beyond a fitted regressor. Jordon et al. (2018) judge synthetic data by whether it preserves the ranking of learning algorithms, a goal close to ours that we measure as paired retention instead of ranks. The phrase *generative compression* already names learned compression of images and video (Santurkar et al., 2017); we use it descriptively, for a table-generating program judged by a declared learning loss, and claim no priority for it.

Evaluation. Train-on-synthetic/test-on-real comparisons originate with Esteban et al. (2017); utility and fidelity measures (Snoke et al., 2018; Raab et al., 2018; Alaa et al., 2022; Zhang et al., 2020) and benchmark suites (Qian et al., 2023; McElfresh et al., 2023; Erickson et al., 2026) broaden them. Data-IQ (Seedat et al., 2022) stratifies training examples by a learner's training dynamics and aleatoric uncertainty and, as a secondary application, compares datasets, synthetic ones included, by those subgroups; it is close evaluation prior art, and its name is why this paper uses no "data IQ" label or intelligence scale. Membership attacks on synthetic data (Shokri et al., 2017; Hayes et al., 2018; van Breugel et al., 2023; Stadler et al., 2022) and holdout-referenced distances (Platzer & Reutterer, 2021) assess privacy empirically, which is distinct from differential privacy (Dwork & Roth, 2014). Our retention is a null-normalized TSTR/TRTR ratio; the new elements are the controls, the paired multi-seed design, and the byte axis.

3 DOPE

3.1 Problem

A table D is split into fit rows D_{fit} and validation rows D_{val}. A compiler maps D_{fit} and a budget B to an artifact A whose charged bytes, every file that sampling reads, are at most B. A sampler draws N rows from A alone. An auditor trained on those rows and scored on D_{val} yields the retention of Section 4.2. The aim is high retention under several auditors at small charged size, with each selection step stated. Three nearby objects keep predictive behavior in another form: a compact predictor keeps $\mathbb{E}[y \mid x]$ but no training rows, which the predictor-only control approximates; a

Algorithm 1 DOPE compile and sample. Training reads the fit rows only; sampling reads the artifact only.

Compile(D_{fit}, B):

1. Fit each feature's marginal and transform T_j and record its missing rate m_j.
2. Keep the twelve features with the largest absolute rank correlation with y; call them S.
3. Train (1) with AdamW (learning rate 0.002, clipped gradients) for the profile's step budget; freeze (b, W) and refit (a, β, v) in closed form.
4. Fit the noise model on the residuals $y - \hat{y}(x)$.
5. For each dependence family, fit it on the Gaussianized features, quantize, encode, and record a proxy score and the gate shortfall.
6. Return the candidate with the smallest gate shortfall, then the fewest bytes; an artifact above the budget B fails the release gate.

Sample(A, N): for each of N rows, draw z from the stored dependence; set $x_j = T_j^{-1}\big(F_j^{-1}(\Phi(z_j))\big)$; set $y = \text{clamp}_{[0,1]}\big(\hat{y}(x_S) + \sigma(x)\,\varepsilon\big)$ with $\varepsilon \sim \mathcal{N}(0,1)$; replace each feature by a missing mark with probability m_j.

coreset or distilled set keeps rows; and a large generator models the joint table with many times the bytes. DOPE keeps a sampling program. The predictor-only control and the generator comparators below test it against the first and third objects, the real-row control shows what extra real rows give, and this paper measures no distilled-set comparator.

3.2 The artifact

A DOPE artifact for a numeric table with features $x \in \mathbb{R}^d$ and target y stores five parts. (i) For every feature, a quantized marginal, a monotone transform onto the unit interval, and a missing-value rate. (ii) A dependence model over the Gaussianized features, chosen among seven families: independence, a Chow–Liu tree, a triangular autoregression, a sparse Gaussian copula, a factor model, a truncated vine, and a two-component mixture. (iii) A residual model for the target on a screened support x_S of at most twelve inputs, ranked by absolute rank correlation with y:

$$h_j = \tanh\big(b_j + w_j^\top x_S\big), \qquad \hat{y}(x) = a + \beta^\top x_S + v^\top h, \tag{1}$$

with sixteen hidden units. (iv) A heteroscedastic noise model: a linear fit of $\log|y - \hat{y}|$ on up to four screened features, with its scale clamped between the lower and upper deciles of the absolute residuals. (v) The positions of the screened columns. The compiler quantizes every part and encodes the artifact; the byte charge is the encoded model file plus the `projection.json` that maps raw columns onto the unit interval. An artifact larger than 10,240 bytes fails the release gate.

3.3 Compile, sample, and profile

Algorithm 1 lists the two procedures. The compile search reads the fit rows only. The reported profile, `features12_steps2048`, trains for 2,048 steps; the untuned trainer default trains for 96 steps with a width between 4 and 12. Four bounded research profiles (twelve or twenty-four screened inputs; 512 or 2,048 steps) were compared by validation retention at fit seed 11. The comparison ran on a six-lineage discovery panel, on six disjoint confirmation lineages, and then on all 100 lineages of the panel below under all three auditors at n and $4n$. Each of those publications keeps all four profiles and selects none, and no contemporaneous record states the rule or the rows that singled out the reported profile (supplement A.2). We therefore treat every lineage and all three auditors as selection-exposed. Section 5.6 and Tables 1 and 2 report the untuned default beside it, and the sealed test removes this selection from the final estimate.

3.4 Release gate

Retention alone cannot decide a release: an artifact can score well and still copy rows, miss the joint distribution, or exceed the byte cap. MFS-v3 combines seven components $c_i \in [0, 1]$ (utility transfer 0.35; representation closeness, driver fidelity, and distribution fidelity 0.15 each; structure fidelity

0.10; coverage realism and compactness 0.05 each) as

$$\text{MFS-v3} = \text{clip}_{[0,100]}\left(100\exp\left(\frac{\sum_i w_i \ln(\max(c_i,\epsilon))}{\sum_i w_i}\right)\right), \quad \epsilon = 10^{-6}, \tag{2}$$

and returns null unless four prerequisites hold (a real-versus-real control, both size cells, artifact-only sampling, locked metric implementations) and every hard gate passes; membership, attribute, and copy failures are gates, so utility cannot offset them. Compactness is $\text{clamp}(1-\text{bytes}/10240, 0, 1)$. A conformance test ties (2) to the contract and the scorer (supplement B). The score is null for every method in this paper because the four prerequisites and the cleartext scan are unmeasured; measured copy, membership, and attribute gates also fail on some method–lineage cells.

4 Protocol

4.1 Data and partitions

The panel holds the 100 rights-cleared PMLB regression lineages (Olson et al., 2017; Romano et al., 2022) that passed the data lock; 65 of them come from four simulated families (Feynman, Strogatz, Friedman-style `fri`, and BNG generators). Large sources are reduced by a fixed coreset to at most 800 fit rows. The official training rows are split $80/20$ by seed 1729 into fit and validation rows, grouping duplicates. The official test rows stay unread until a single pre-registered run; every number here is a validation estimate.

4.2 Retention

Let L_{null}, L_{real}, and L_{syn} be an auditor's validation mean squared error, on the target min–max projected to $[0,1]$ by the training split, when it predicts the training mean, when it is trained on the real fit rows, and when it is trained on synthetic rows. Retention is

$$R = \frac{L_{\text{null}} - L_{\text{syn}}}{L_{\text{null}} - L_{\text{real}}}, \tag{3}$$

so $R = 1$ matches the real-trained auditor and $R < 0$ is worse than the null. A lineage is informative for an auditor when $L_{\text{null}} - L_{\text{real}} \geq 0.01\,|L_{\text{null}}|$. The task-trained auditors are CatBoost (Prokhorenkova et al., 2018), ordinary least squares, and a one-hidden-layer MLP; CatBoost and the MLP use seed 1729. On the density block we also score three pretrained in-context tabular foundation models—NVIDIA Kumo large (`kumo_tabular_l`), AutoGluon Mitra (`mitra_v2`), and TabICL (`tabicl2`)—where the synthetic rows replace the real fit rows as in-context conditioning examples without weight updates (supplement B). A synthetic table has n or $4n$ rows, where n is the fit-row count, and three sample seeds. A fit's value is the median over sample seeds; a lineage's value is the median over its fits.

4.3 Comparators and controls

Comparators are author-default and native-selected ARF (Watson et al., 2023) at five fit seeds; TabSyn with a scaled schedule of 200 VAE and at most 1,000 diffusion epochs, which the authors' early stop can end sooner, at five fit seeds; Gaussian copula, Chow–Liu, and independent marginals selected by held-out log density; CTGAN and TVAE selected on validation skill of the linear and MLP auditors; and a historical Forest-Flow block. Supplement C lists each method's selection rule, data, and budget. TabSyn's five-seed panel has two gaps: six sample cells of one seed-23 lineage reached the sample time cap and stay unscored, and one other lineage has no stored seed-11 samples. Every lineage keeps at least four complete fit seeds, so all enter the five-seed comparison, and TabSyn's CatBoost retention row at $4n$ rests on 99 informative lineages. The *predictor-only* control fits a ridge regression on the fit rows, samples each feature independently from its marginal, and adds Gaussian residual noise; it keeps a regressor and drops the joint model. The *real-row* control resamples the fit rows with replacement to $4n$ rows; it is a diagnostic of what extra rows alone do, neither a generator nor an upper bound on retention.

4.4 Statistics

Comparisons pair lineages that both arms complete (a five-seed arm needs three complete fits). The paired estimand is the Hodges–Lehmann pseudo-median of the lineage differences, the location the Wilcoxon signed-rank test addresses. Intervals come from a nested bootstrap with $10{,}000$ draws: clusters (each simulated family is one cluster; every other lineage is its own), then lineages inside a drawn cluster, then fit seeds inside a drawn lineage, separately for each arm. Wilcoxon p-values are Holm-adjusted inside the pre-declared families of each size (Table 1). The Wilcoxon test treats lineages as independent; each verdict reads the clustered interval. A 90% interval inside ± 0.02 would establish equivalence. No generator comparison meets that rule. These choices were committed before the five-seed refit ran, but after an earlier four-seed run whose samples the refit reproduces at seeds 23 to 71 (Section 5.6).

5 Results

5.1 DOPE against the strongest comparators across model families

Table 1 gives the paired differences at $4n$ across all three task-trained auditors, while Fig. 1 places each method's retention against its median charged bytes and cross-architecture profile. At $4n$, DOPE reaches median retention 1.133 (1.029 to 1.246) under the neural MLP, 0.991 (0.972 to 0.997) under linear regression, and 0.898 (0.804 to 0.972) under CatBoost. Under the linear auditor, DOPE exceeds TabSyn (scaled schedule) by 0.069 (0.005 to 0.210) and author-default ARF by 0.062 (-0.011 to 0.193), and against the Gaussian copula the difference is 0.007 (-0.023 to 0.030). Under the neural MLP, DOPE exceeds author-default ARF by 0.137 (0.041 to 0.397), the Gaussian copula by 0.234 (0.126 to 0.390), and TabSyn (scaled schedule) by 0.054 (-0.006 to 0.277). Under CatBoost, the strongest full-panel comparator is TabSyn (scaled schedule): DOPE cannot be separated from it, with difference -0.003 (-0.053 to 0.053; $n = 99$; W/T/L 53/0/46). Against author-default ARF the CatBoost difference is 0.048 (-0.034 to 0.149; $n = 99$), against the Gaussian copula it is 0.191, and against Chow–Liu 0.356. No generator comparison is equivalent under the pre-declared rule; "cannot be separated from" means that the interval contains zero, not that the two methods are interchangeable. TVAE ($n = 21$) and CTGAN ($n = 21$) are small cohorts drawn with open marks; their tested author-default configurations are in supplement E. On the 6 lineages of the historical Forest-Flow block, DOPE is below Forest-Flow (-0.231, -0.382 to -0.005; W/T/L 0/0/6), whose median artifact is $325.8\,\mathrm{MB}$; supplement E gives every row.

5.2 What the controls show

The predictor-only control keeps a fitted ridge regressor and drops the joint feature model, so its gap bundles DOPE's dependence model with DOPE's nonlinear residual. Under CatBoost, DOPE minus predictor-only is 0.197 (0.112 to 0.267); under the MLP it is 0.227 and under the linear auditor 0.008. A tree or network auditor trained on independently drawn features sees feature combinations that the real table never contains, while a linear auditor depends mostly on the regressor. The real-row control resamples real rows to $4n$; DOPE minus that control is -0.179 (-0.276 to -0.074) under CatBoost and -0.037 under the MLP, so DOPE is below resampled real rows under CatBoost. Extra rows lift an MLP auditor even when they are copies, which is why MLP retentions above one appear.

5.3 Both synthetic sizes

Table 2 reports retention levels at n and $4n$ for every arm; the two sizes are co-primary in the original pre-registration, the v2 contract places $4n$ in the headline, and $4n$ gives each auditor four times as many synthetic rows as the real fit view. Moving from n to $4n$ takes DOPE's CatBoost median from 0.824 to 0.898 and raises the CatBoost median of 6 of 6 full-panel comparators; the CatBoost ranking of the full-panel arms is the same at both sizes. Resampling the real rows to $4n$ gives 1.018 under CatBoost and 1.174 under the MLP, which is what row count alone does to each auditor.

Table 1: **DOPE minus each comparator at synthetic size $4n$.** Hodges–Lehmann paired difference in retention with its 95% nested bootstrap interval for each auditor; n, W/T/L (DOPE higher, equal, lower), and the Holm-adjusted p-value are for CatBoost. Holm families are pre-declared per size: the primary family has 12 tests (author-default ARF, TabSyn, TabDDPM, and Forest-Flow under the common budget, by three auditors), the density family 9 (Gaussian copula, Chow–Liu, and independent marginals, by three auditors), a deep-generator family 12 (author-default CTGAN and TVAE, TabDiff, GReaT), a secondary family 9 (native-selected ARF, synthpop, SMOTE), and the control family 6 (predictor-only control and resampled real rows, by three auditors); untested members keep their slot. A dash marks a descriptive row outside every family; the tested author-default CTGAN and TVAE rows are in supplement E. Bytes are the median charged artifact bytes over each method's lineages with a charged fit, with that lineage count beneath; for TVAE and CTGAN it exceeds the scored n.

Comparator	n	CatBoost	Linear	MLP	W/T/L	Holm p	Bytes
TabSyn (scaled schedule)	99	-0.003 $[-0.053, 0.053]$	0.069 $[0.005, 0.210]$	0.054 $[-0.006, 0.277]$	53/0/46	1.000	42.3 MB (100)
ARF (author default)	99	0.048 $[-0.034, 0.149]$	0.062 $[-0.011, 0.193]$	0.137 $[0.041, 0.397]$	60/0/39	0.229	1.4 MB (100)
ARF (native selected)	99	0.083 $[0.006, 0.191]$	0.083 $[0.006, 0.217]$	0.191 $[0.060, 0.431]$	66/0/33	0.004	1.6 MB (100)
Gaussian copula	99	0.191 $[0.089, 0.273]$	0.007 $[-0.023, 0.030]$	0.234 $[0.126, 0.390]$	83/0/16	1.46×10^{-13}	5.5 KB (100)
Chow-Liu tree	99	0.356 $[0.242, 0.728]$	0.495 $[0.266, 0.816]$	0.658 $[0.435, 1.166]$	91/0/8	5.59×10^{-15}	29.9 KB (100)
Independent marginals	99	0.881 $[0.790, 0.964]$	0.982 $[0.926, 1.008]$	1.327 $[1.143, 1.629]$	99/0/0	5.13×10^{-17}	2.5 KB (100)
TVAE (native selected)	21	0.201 $[0.097, 0.339]$	0.262 $[0.145, 0.473]$	0.426 $[0.189, 1.231]$	19/0/2	—	308.6 KB (34)
CTGAN (native selected)	21	0.912 $[0.731, 1.298]$	1.066 $[0.812, 1.695]$	1.439 $[0.788, 2.272]$	21/0/0	—	731.0 KB (35)
Forest-Flow (historical)	6	-0.231 $[-0.382, -0.005]$	0.002 $[-0.051, 0.117]$	-0.080 $[-0.562, 0.004]$	0/0/6	—	325.8 MB (6)
DOPE, untuned default	99	0.134 $[0.053, 0.208]$	-0.003 $[-0.026, 0.005]$	0.138 $[0.052, 0.251]$	87/0/12	—	1.7 KB (100)
Predictor-only control	99	0.197 $[0.112, 0.267]$	0.008 $[-0.027, 0.046]$	0.227 $[0.118, 0.378]$	92/0/7	2.80×10^{-14}	—
Real rows resampled	99	-0.179 $[-0.276, -0.074]$	-0.008 $[-0.046, 0.008]$	-0.037 $[-0.101, 0.070]$	10/0/89	2.02×10^{-13}	—

5.4 Utility per byte, foundation models, and driver fidelity

Fig. 1 panel (a) places each method's retention at $4n$ against its median charged bytes. DOPE's median artifact is 1,895 bytes, and $490/500$ fits are within 10,240 bytes; the 10 fits above it are every fit of 4544_GeographicalOriginalofMusic and 588_fri_c4_1000_100 (19,019 to 22,233 bytes). Without those 2 lineages, a sensitivity added after review, DOPE's CatBoost median at $4n$ is 0.898 (0.803 to 0.975; $n = 97$), and DOPE cannot be separated from TabSyn (scaled schedule) (-0.008, -0.060 to 0.050; $n = 97$). The strongest full-panel comparator's median artifact is 42.3 MB, and author-default ARF's is 1.4 MB. Over every measured generator, including DOPE's untuned default, the Pareto set of median bytes and CatBoost retention is {DOPE (ours); DOPE, untuned default; TabSyn (scaled schedule); Forest-Flow (historical)}. The points are unpaired medians that order methods without testing them, and a hollow point rests on fewer than 90 lineages; Table 1 gives the paired comparison. The median artifacts of the neural, diffusion, and forest generators are 1.6×10^2 to 1.7×10^5 times DOPE's. On the 97-lineage density block (Table 3, Fig. 1 panel b), DOPE also preserves context for three pretrained in-context tabular foundation models without fine-tuning, reaching median retention 0.863 [0.809, 0.880] under NVIDIA Kumo large, 0.871 [0.835, 0.893] under TabICL, and 0.858 [0.803, 0.887] under AutoGluon Mitra, and exceeds the Gaussian copula, Chow–Liu, and independent marginals under all three foundation models. On the 94 informative lineages of the same block, holdout feature driver fidelity—comparing real-trained and synthetic-trained permutation-importance rankings on held-out real rows—reaches median 0.905 (interquartile range 0.737 to 0.948) for DOPE, against 0.923 (0.728 to 0.955) for the Gaussian copula, 0.775 (0.562 to 0.899) for Chow–Liu, and 0.304 (0.145 to 0.491) for independent marginals (supplement B).

Table 2: **Retention level by method, auditor, and synthetic size.** Median over lineages with the 95% nested bootstrap interval beneath it; n is the CatBoost lineage count of that arm, and the linear and MLP counts, which drop lineages where that auditor is uninformative, are in supplement E. Five-seed arms use the median of at least three complete fits per lineage. Controls are references, not generators. No Holm family.

		CatBoost		Linear		MLP	
Method	n	n rows	$4n$ rows	n rows	$4n$ rows	n rows	$4n$ rows
DOPE (ours)	99	0.824 [0.73, 0.94]	0.898 [0.80, 0.97]	0.986 [0.96, 0.99]	0.991 [0.97, 1.00]	0.969 [0.90, 0.99]	1.133 [1.03, 1.25]
DOPE, untuned default	99	0.636 [0.39, 0.82]	0.704 [0.42, 0.88]	0.986 [0.97, 0.99]	0.993 [0.98, 1.00]	0.887 [0.80, 0.96]	0.991 [0.96, 1.06]
TabSyn (scaled schedule)	99	0.863 [0.75, 0.93]	0.924 [0.81, 0.96]	0.932 [0.79, 0.98]	0.952 [0.83, 0.99]	0.884 [0.73, 0.98]	1.013 [0.95, 1.14]
ARF (author default)	99	0.718 [0.57, 0.84]	0.822 [0.69, 0.89]	0.903 [0.79, 0.96]	0.938 [0.85, 0.98]	0.828 [0.63, 0.89]	0.991 [0.89, 1.09]
ARF (native selected)	99	0.671 [0.55, 0.83]	0.804 [0.65, 0.89]	0.894 [0.74, 0.97]	0.920 [0.83, 0.98]	0.805 [0.58, 0.90]	0.966 [0.84, 1.03]
Gaussian copula	99	0.610 [0.31, 0.78]	0.669 [0.37, 0.87]	0.961 [0.91, 0.99]	0.987 [0.96, 1.00]	0.800 [0.62, 0.89]	0.978 [0.89, 1.07]
Chow-Liu tree	99	0.466 [0.25, 0.58]	0.576 [0.29, 0.64]	0.636 [0.27, 0.76]	0.659 [0.28, 0.76]	0.454 [0.18, 0.63]	0.660 [0.35, 0.80]
Independent marginals	99	−0.046 [−0.08, −0.03]	−0.012 [−0.04, 0.00]	−0.050 [−0.08, −0.01]	0.002 [−0.03, 0.02]	−0.215 [−0.48, −0.07]	−0.049 [−0.16, −0.02]
TVAE (native selected)	21	0.606 [0.41, 0.74]	0.674 [0.45, 0.78]	0.764 [0.55, 0.86]	0.796 [0.58, 0.88]	0.721 [0.25, 0.80]	0.812 [0.64, 0.88]
CTGAN (native selected)	21	−0.074 [−0.48, 0.12]	−0.080 [−0.28, 0.22]	−0.056 [−0.38, 0.12]	−0.033 [−0.45, 0.12]	−0.247 [−1.07, 0.10]	0.033 [−0.34, 0.33]
Forest-Flow (historical)	6	0.977 [0.81, 1.00]	1.011 [0.91, 1.04]	0.981 [0.75, 1.08]	0.985 [0.98, 1.00]	0.885 [0.58, 1.00]	1.054 [0.93, 2.73]
Predictor-only control	99	0.596 [0.34, 0.73]	0.631 [0.38, 0.78]	0.970 [0.92, 0.99]	0.981 [0.95, 1.00]	0.849 [0.73, 0.91]	0.947 [0.88, 1.02]
Real rows resampled	99	—	1.018 [1.00, 1.03]	—	0.998 [0.99, 1.00]	—	1.174 [1.05, 1.23]

Table 3: **Zero-shot tabular foundation models (Kumo large, Mitra, and TabICL) on the density block at $4n$.** Each cell is a retention median and 95% bootstrap interval; the denominator is the paired informative lineages in that row. Holm family size is three per auditor: that auditor's DOPE against the Gaussian copula, Chow–Liu, and independent marginals. W/T/L counts lineages; a tie is a difference of zero.

Auditor	Comparator	n	DOPE	Other	Difference	W/T/L	Holm p
Kumo large	Gaussian copula	97	**0.863** [0.809, 0.880]	0.747 [0.668, 0.778]	0.059 [0.0098, 0.116]	62/0/35	5.70×10^{-4}
Kumo large	Chow–Liu	97	**0.863** [0.809, 0.880]	0.562 [0.501, 0.629]	0.247 [0.195, 0.306]	85/0/12	1.34×10^{-11}
Kumo large	Indep. marginals	97	**0.863** [0.809, 0.880]	-0.000819 [-0.00189, 7.58e-05]	0.874 [0.834, 0.888]	94/0/3	7.46×10^{-17}
Mitra	Gaussian copula	97	**0.858** [0.803, 0.887]	0.713 [0.585, 0.771]	0.066 [0.023, 0.135]	65/0/32	1.34×10^{-4}
Mitra	Chow–Liu	97	**0.858** [0.803, 0.887]	0.583 [0.527, 0.643]	0.242 [0.182, 0.293]	85/0/12	2.71×10^{-11}
Mitra	Indep. marginals	97	**0.858** [0.803, 0.887]	-0.00907 [-0.014, -0.00377]	0.890 [0.840, 0.909]	94/0/3	6.01×10^{-17}
TabICL	Gaussian copula	97	**0.871** [0.835, 0.893]	0.754 [0.672, 0.787]	0.079 [0.00947, 0.116]	62/0/35	1.61×10^{-4}
TabICL	Chow–Liu	97	**0.871** [0.835, 0.893]	0.587 [0.484, 0.650]	0.267 [0.200, 0.302]	87/0/10	7.12×10^{-12}
TabICL	Indep. marginals	97	**0.871** [0.835, 0.893]	-0.00446 [-0.012, 0.0012]	0.891 [0.846, 0.919]	92/0/5	5.83×10^{-17}

Holm family: 3 tests per auditor; DOPE versus Gaussian copula, Chow–Liu, and independent marginals.

5.5 Fidelity and empirical privacy

Table 4 reports detection and distance measures at $4n$ on the fit-seed-11 samples of DOPE, ARF, the density baselines, TVAE, and CTGAN; TabSyn's distance and membership cells belong to the pre-declared privacy block. Forest-Flow's stored sample files support its retention re-score, but the privacy and detector runs record its cells as unavailable. A CatBoost two-sample detector, trained on at most 200 rows per class because the validation cut is that small, separates DOPE rows from validation rows with median AUC 0.640 (0.583 to 0.831), so the rows are distinguishable to a strong classifier even where an auditor's retention is high. The median distance from a DOPE row to its nearest fit row is 0.080 and to its nearest validation row 0.109; the fit view holds about four times as many rows as the validation cut, so the first distance is smaller even for a fresh draw. A distance-based membership attack reaches median AUC 0.505 (0.492 to 0.516). On the five-seed cells, DOPE's mean marginal Kolmogorov–Smirnov distance is 0.117 and its pairwise correlation fidelity

Table 4: **Detection and empirical privacy at $4n$, fit-seed-11 samples.** Medians over lineages with 95% family-cluster intervals; TabSyn's distance cells are pending, and the privacy run records Forest-Flow's cells as unavailable. C2ST is a CatBoost two-sample AUC (one half is indistinguishable). DCR fit and DCR val. are median distances to the nearest fit row and to the nearest validation row; the fit reference is about four times larger. NNDR is the nearest-neighbour distance ratio against the fit rows. MIA is a distance-attack AUC. n is the lineage count, given as a range where the metrics differ; a lineage counts when all three of its sample files resolve at the requested row count, a different rule from the informative-retention rule behind the n of Table 1. No Holm family.

Method	n	C2ST	DCR fit	DCR val.	NNDR	MIA AUC
DOPE (fit seed 11)	98	0.640 [0.58, 0.83]	0.080 [0.04, 0.13]	0.109 [0.05, 0.15]	0.905 [0.85, 0.94]	0.505 [0.49, 0.52]
ARF (author default)	100	0.612 [0.57, 0.69]	0.082 [0.03, 0.13]	0.109 [0.05, 0.16]	0.907 [0.86, 0.93]	0.556 [0.54, 0.58]
ARF (native selected)	100	0.609 [0.58, 0.71]	0.083 [0.04, 0.13]	0.111 [0.05, 0.16]	0.911 [0.86, 0.93]	0.541 [0.53, 0.56]
Gaussian copula	99–100	0.735 [0.69, 0.87]	0.086 [0.06, 0.13]	0.107 [0.08, 0.16]	0.920 [0.89, 0.96]	0.502 [0.49, 0.52]
Chow-Liu tree	99–100	0.706 [0.59, 0.94]	0.108 [0.09, 0.17]	0.129 [0.11, 0.19]	0.924 [0.89, 0.96]	0.520 [0.50, 0.54]
Independent marginals	100	0.858 [0.76, 0.97]	0.113 [0.09, 0.16]	0.132 [0.10, 0.18]	0.945 [0.91, 0.97]	0.508 [0.49, 0.52]
TVAE (native selected)	22	0.731 [0.70, 0.89]	0.080 [0.05, 0.15]	0.112 [0.09, 0.19]	0.906 [0.85, 0.93]	0.532 [0.49, 0.54]
CTGAN (native selected)	22	0.954 [0.85, 0.98]	0.153 [0.09, 0.19]	0.187 [0.12, 0.22]	0.941 [0.91, 0.96]	0.502 [0.47, 0.52]

0.953, against 0.103 and 0.962 for author-default ARF and 0.126 and 0.958 for TabSyn. These are empirical measurements and give no guarantee of differential privacy or HIPAA de-identification. Supplement G gives the attack definitions and, for every method with scored samples including TabSyn, the CatBoost detector, marginal and pairwise fidelity, and α-precision and β-recall; Forest-Flow has marginal and pairwise fidelity only.

5.6 Seeds, profile selection, and sensitivity

The first version of this paper reported the historical fit-seed-11 population, with CatBoost median 0.940 at $4n$ when each sample seed also seeded its auditor; re-scored with auditor seed 1729, the same samples give 0.949. Four further seeds, run later with a newer binary, gave 0.862 on 99 lineages. The two binaries write byte-identical seed-11 artifacts on 100 of 100 lineages and byte-identical samples for seeds 23 to 71 on 400 of 400 fits, and the refit reproduces the historical seed-11 samples on 86 of 98 lineages. On the 97 lineages common to every cell, the historical binary gives 0.949 at seed 11 and 0.862 at seeds 23 to 71, and the later binary gives 0.949 and 0.862. Identical artifacts and samples leave the seed, not the binary, as the source of the gap. Seed 11 ranks first among the five seeds on 47 of 99 lineages, against 19.8 if the seeds were exchangeable. Because a seed acts on every lineage at once, as explained next, the lineages are not independent trials, so this count is descriptive and carries no p-value. Seeds 23 to 71 alone give 0.862 (0.771 to 0.958). A fit seed fixes the residual network's initial weights, which are the same on every lineage with the same number of screened inputs, so one seed acts on the whole panel at once; seed 11 is also the only seed at which the four profiles were compared. The five-seed median in the abstract is the estimate this paper stands on. The untuned trainer default gives 0.704 under CatBoost at $4n$, and the reported profile is above it (0.134, 0.053 to 0.208), a difference that includes whatever the profile selection added. Supplement E gives every diagnostic cell and repeats the source-family sensitivity on the five-seed panel: outside the four simulated families, DOPE cannot be separated from TabSyn (scaled schedule) under CatBoost at $4n$ (-0.013, -0.078 to 0.051; $n = 34$) and cannot be separated from author-default ARF (0.015, -0.069 to 0.081; $n = 34$). In the pre-declared all-five-fits sensitivity, which keeps a lineage only where every fit seed completed, DOPE cannot be separated from TabSyn (scaled schedule) (-0.008, -0.059 to 0.048; $n = 97$).

5.7 Multi-level landmark expansion and convergence

Beyond the 100-lineage regression panel above, the repository's multi-level landmark study (supplement H) evaluates `DOPE-MLC-v2` across 1,000 selection-exposed datasets (65,874 downstream fits across 6 auditor families: CatBoost, a local gated-attention network trained from scratch per table, LightGBM, XGBoost, MLP, and linear/logistic) and `DOPE-MLC-v3` across 1,069 datasets (1,000 shared with `v2` plus 69 multiclass tables, 4,276 completed dataset-tier cells across 2, 4, 16, and 64 KiB caps). Across 8 shard-based training rounds (1.333351× effective passes on 18,395 training splits with 684 validation splits, 3.59% $<$ 5%), full validation loss drops by 16.52%

from 0.40988 ± 0.01767 to 0.34216 ± 0.01621 (strictly disjoint 1,000-dataset track: 0.60691 to 0.57363 ± 0.02131; `v3` track: 0.69727 to 0.68492 ± 0.03877). On the 1,000-dataset tournament (44,369 pairwise matches per method), `DOPE-MLC-v2` ranks first at Bradley–Terry Elo 1651.3 [1643.4, 1659.6] (70.58% win share; strictly disjoint 1,000-dataset Elo 1650.7), ahead of raw row subsampling (1593.3), the v1 kernel (1592.5), Gaussian copula (1545.8), joint diffusion (1432.4), and ARF (1184.7), with mean holdout retention 0.8620 at 1,502 B, 0.8971 at 2,836 B, and 0.8975 at 5,872 B, plus bit-exact `SafeTensors` and wire replay in pure Rust (`dope mlc-v3`).

6 LIMITATIONS

Five confounds. (i) *Row count:* retention at $4n$ mixes fidelity with row count; the real-row control diagnoses that effect but does not bound it. (ii) *Predictor-only:* the predictor-only gap credits DOPE's dependence model and its nonlinear residual together. (iii) *Selection on the scoring rows:* validation rows informed DOPE's profile (by retention), the CTGAN and TVAE configurations (by auditor skill), and the native-selected density and ARF configurations (by held-out log density), and the same rows score retention; DOPE's profile comparison read every lineage and all three auditors, and the twelve discovery and confirmation lineages include all four simulated families and three lineages outside them, so neither the auditors nor the source-family sensitivity is held out from that choice, and the sealed test run is the only estimate free of it. (iv) *Lineage dependence:* 65 of 100 lineages come from four simulated families, and a DOPE fit seed acts on every lineage at once; clustering by family widens the intervals but cannot add independent sources, and the nested interval, which resamples seeds within each lineage, does not widen for that shared effect. (v) *Single fit seed:* DOPE, ARF, and TabSyn have five fit seeds, but the density baselines, CTGAN, and TVAE contribute one fit seed each, so their side of each interval omits fitting variance. **Scope.** The panel is regression only, and fit views hold at most 800 rows. **Baselines.** TabSyn runs a scaled schedule of 200 VAE and at most 1,000 diffusion epochs against the authors' 4,000 and 10,001, TVAE and CTGAN cover small cohorts, and Forest-Flow is a historical block of 6 lineages. The pre-declared five-seed density runs, synthpop, SMOTE, Forest-Flow under the common budget, TabDDPM, TabDiff, and an LLM generator are pending. **Privacy.** The privacy rows are empirical attacks on released rows; attacks on the artifact itself and shadow-model attacks are pending.

7 CONCLUSION

Downstream Objective-Preserving Encoding (D.O.P.E., or DOPE) compiles a regression table into a sampling program whose median charge is 1,895 bytes, and asks how much of the table's predictive structure diverse learners recover from it. At $4n$ it retains 1.133 of a real-trained auditor's gain under the neural MLP, 0.991 under linear regression, and 0.898 under CatBoost, transfers context across pretrained tabular foundation models (Kumo large, TabICL, and Mitra) on the density block, cannot be separated from TabSyn (scaled schedule) at about 2.2×10^4 times fewer bytes, and is above a predictor-only sampler. The release score is null because its prerequisites and cleartext scan are unmeasured, and measured copy, membership, and attribute gates already fail on some cells. The official test split waits for one pre-registered run. The analysis contract, every ledger, and the scripts that regenerate each number from those ledgers are public, so each figure and table in this paper can be rebuilt from the repository with one command.

AVAILABILITY

The analysis contract (`predeclare_v2.json`) was committed before the five-seed refit ran, after an earlier four-seed run, and a continuous-integration job deletes every generated input and rebuilds the paper from the committed ledgers. Jankurai is the repository's conformance auditor. A passing audit means the build, tests, and boundary proofs cleared the release floor with no high finding. It is not a utility score, and it is separate from any MFS value and from any privacy certificate. The implementation and the receipts behind these tables are at

https://github.com/neverhuman/dope.

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